Lab 02 - Regression

1. Introduction

This project is about predicting the song popularity using audio features and metadata. Two approaches were used: Submission 1 combined Linear Regression with boosting models, while Submission 2 used an optimized LightGBM pipeline with advanced encoding. Performance was evaluated using RMSE along with MAE, R^2, and MAPE.

2. Dataset

- The dataset consists of three files: train.csv, test.csv, sample_submission.csv
- Each row represents a music track containing audio features (such as energy, tempo, and danceability), derived features (such as emotional charge and groove efficiency), and metadata (such as release date and artist details).
- **Dataset Details:** training samples: 61,609, test samples: 41,074, features: 61
- **Target Variable:** Continuous popularity score (0–100)

Additional details: https://www.kaggle.com/competitions/mlx-2-0-regression/data

3. Data preprocessing

Submission 1:

Performs feature engineering on the dataframe, including converting publication_timestamp into year, month, day, and weekday features. Aggregated and interaction features are created, and identifier/label columns are removed. Numeric values are median-imputed, categorical values are filled with "Unknown", and categorical features are one-hot encoded using a ColumnTransformer.

Submission 2:

Uses a more advanced preprocessing approach by removing track_identifier, extracting year and month from publication_timestamp, and applying frequency encoding to composition labels. Missing values are handled using median (numeric) and "Unknown" (categorical). Categorical features are encoded using ordinal encoding (low-cardinality) and target encoding (high-cardinality), along with additional aggregated and interaction features before training a LightGBM model.

4. Models Used

In Submission 1, Linear Regression was used as a baseline because it is simple, fast, and provides a reference point for performance. However, it assumes a linear relationship between features and the target, which limits its ability to capture complex patterns in the data. To address this, boosting models such as CatBoost, XGBoost, and LightGBM were used, as they build ensembles of decision trees that can model non-linear relationships and feature interactions.

In Submission 2, LightGBM was selected as the primary model due to its efficiency and strong performance on tabular data. It uses histogram-based learning and leaf-wise tree growth, which allows it to capture complex structures while remaining computationally efficient. However, these models require careful tuning and are less interpretable compared to linear models.

Boosting models work by sequentially training decision trees, where each new tree focuses on correcting the errors made by previous ones. This iterative process improves model accuracy and allows the model to learn complex relationships in the data.

Model	**Description**	**Why Chosen**	**Advantages**	**Disadvantages**
Linear Regression (Submission 1 Baseline)	A baseline regression model that predicts popularity as a weighted linear combination of input features after preprocessing.	Used as a fast and interpretable benchmark to compare against more complex models.	Very fast training and inference, simple to interpret, low computational cost.	Cannot capture non-linear relationships or complex feature interactions, leading to lower accuracy on rich datasets.

Advanced Boosting Models (Submission 1: CatBoost, XGBoost, LightGBM, Gradient Boosting + Blending)	Ensemble models that build boosted decision trees; multiple models are compared and optionally blended for better performance.	Chosen to capture non-linear patterns and feature interactions and improve leaderboard performance.	High predictive power, handles tabular data well, strong performance in competitions.	More complex to tune, higher training time, less interpretable, risk of overfitting.
LightGBM Focused Pipeline (Submission 2)	A single optimized LightGBM model trained on engineered and encoded features (frequency, ordinal, target encoding).	Selected for a strong balance between accuracy and efficiency on tabular data.	Fast training, scalable, high accuracy with proper feature engineering.	Sensitive to hyperparameters and preprocessing, less interpretable than simple models.

5. Evaluation Metrics

1. Root Mean Squared Error (RMSE): Main competition metric, Penalizes large errors
2. Mean Absolute Error (MAE): Average absolute difference between predictions and actual values
3. R^2 Score: Measures variance explained by the model
4. Mean Absolute Percentage Error (MAPE): Measures relative error percentage

Lower RMSE and MAE values in Submission 2 indicate improved prediction accuracy and reduced average error. The higher R^2 score (0.78 vs 0.67) shows that the LightGBM model explains more variance in the data. Additionally, the consistency between validation and leaderboard scores suggests better generalization compared to Submission 1.

6. Results

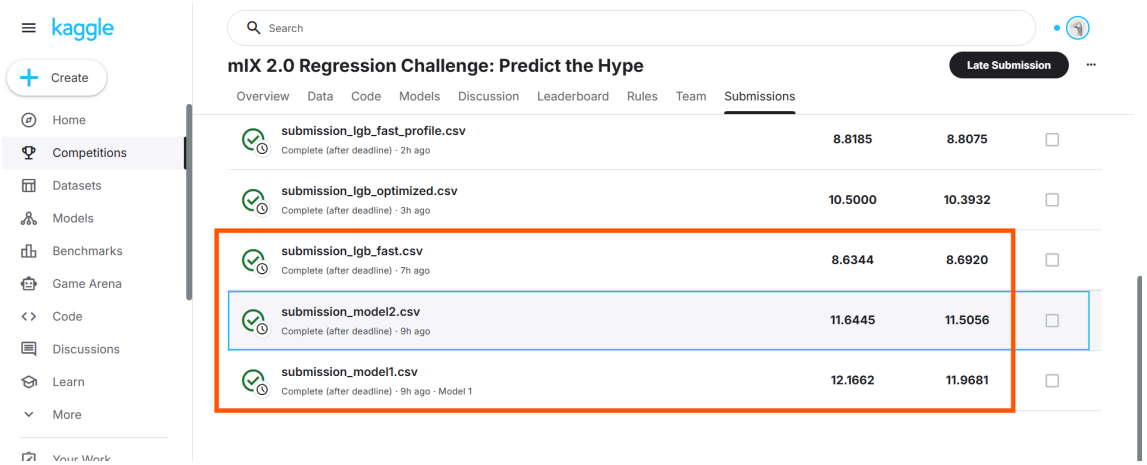
Validation Performance

Submission	Model	RMSE	MAE	R^2	Notes
Submission 1	Linear Regression	12.41	8.18	0.67	Baseline model
Submission 1	CatBoost	11.75	8.55	0.70	Improved over baseline
Submission 1	Blended Model	11.49	–	–	Best validation result in Submission 1
Submission 2	LightGBM	10.08	5.43	0.78	Best overall validation performance

Kaggle Leaderboard Results

Submission	File Name	Private Score (RMSE)	Public Score (RMSE)
Submission 1	submission_model1.csv	12.17	11.97
Submission 1	submission_model2.csv	11.64	11.51
Submission 2	submission_lgb_fast.csv	8.63	8.69

7. Conclusion

Submission 2 achieved the best performance, with lower RMSE and MAE indicating higher accuracy and a higher R^2 showing better explanatory power. The close match between validation and leaderboard results suggests good generalization. This improvement is mainly due to LightGBM and more effective preprocessing and encoding, which capture non-linear relationships better.

8. Screenshot of the submission